

Azizullah Khan

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EDUCATION

Institute of Business Administration, Karachi

Aug 2021 – May 2025

Bachelor of Science in Computer Science | CGPA: 3.76 (Dean's List)

PROFESSIONAL EXPERIENCE

Data Science Dojo. Software Engineer - I (Infrastructure)

Aug 2025 – Present

- Designed a zero-trust hub-spoke network topology across **8 spokes**, privatizing all stateful resources as the first phase toward full infrastructure privatization, reducing attack surface and contributing to SOC2, GDPR, and HIPAA compliance.
- Enforced compliance standards at the infrastructure layer through policy-driven automation, ensuring every deployed resource is validated at deployment time and automatically remediated, eliminating manual compliance checks.
- Deployed Langfuse on Azure Kubernetes Service (AKS) to enable LLM observability for internal AI/ML workloads, provisioning ClickHouse, PostgreSQL, and Azure Blob Storage privately with no public endpoints, keeping prompt traces and model metadata inside the compliance boundary and enabling self-hosted control over the observability stack.
- Developed modular Infrastructure-as-Code reducing environment provisioning time from **~2 hours** to **~20 minutes**, supporting **8 environments** with consistent, repeatable infrastructure setup.
- Standardized CI/CD pipelines across services with caching, parallelization, and SonarQube quality gates, significantly reducing pipeline runtime and enforcing test coverage at the build stage.
- Owned end-to-end design and delivery of a multi-tenant billing service, replacing a manual billing process with automated event-driven billing, unlocking custom on-demand usage tracking across **8 billing dimensions**, proactive usage alerts, and support for all billing cadences across **5 tenants**.
- Built a centralized monitoring service achieving **95%+** tenant test coverage, reducing incident detection time from **~1 hour** to **8 minutes** and proactively catching **10+** potential incidents before customer impact.
- Established production-grade on-call practices, onboarding engineers across teams with structured playbooks and training, enabling independent incident investigation and rapid reporting.

Salsoft Technologies. Junior Backend Developer

Jan 2024 – Jul 2025

- Developed backend services for three production applications using modular, testable code and structured code reviews, supporting consistent feature delivery and long-term maintainability.
- Implemented core backend features including role-based access control, real-time notifications, scalable chat systems, and advanced search with dynamic filtering and sorting.
- Optimized high-traffic APIs by adding database indexes, rewriting inefficient queries, and introducing caching, achieving **~5%** performance improvement while contributing to architecture discussions, codebase structuring, and database schema design.

PROJECTS

Indus Sahulat - Emergency Healthcare Response System (Final Year Project, Spring Boot, PostgreSQL)

- Developed a real-time emergency healthcare platform improving coordination between patients, hospitals, and drivers, reducing ambulance response time **~10%** through efficient dispatch and alert systems.
- Designed secure, scalable and modular REST APIs to handle emergency alerts and dispatch workflows, improving reliability and minimizing failures under load through proper validation, error handling, and optimized backend logic.
- Integrated Spring Security with fine-grained role-based access control for patients, ambulance providers, drivers, and admins, enhancing data protection and securing critical workflows across the platform's multi-user environment.
- Enabled WebSocket-based real-time tracking and Redis-backed location caching, reducing response latency and decreasing write operations to the database by **~20%** during peak emergency loads.

AWS Cloud Infrastructure Deployment (AWS, CI/CD)

- Designed and deployed a highly available 3-tier architecture on AWS across multiple availability zones, with public and private subnets, an internet-facing load balancer, and autoscaling for the application layer.
- Implemented a Jenkins CI/CD pipeline for automated build and deployment to EC2 instances, with custom domain routing via Route 53.

SKILLS

Cloud Platforms: Azure | AWS

Infrastructure & DevOps: Linux | Docker | Kubernetes | Terraform | CI/CD Pipelines

Observability: Prometheus | Grafana | Azure Monitor | PagerDuty

Languages: Python | Java | JavaScript | TypeScript

Frameworks: FastAPI | Spring Boot | Node.js | React.js

Database Systems: PostgreSQL | MongoDB | Redis

CERTIFICATIONS

Microsoft Azure Certifications – AZ-900 | AZ-204 | AZ-140 | AZ-400 (Devops Engineer Expert)

NVIDIA - Fundamentals of Accelerated Computing with CUDA Python